

ANALOG ELECTRONICS

ARNO, CHANG-SHAO, EZGI, XUEPU

STRICTLY CONFIDENTIAL

SUMMARY

- Session 1:
  - How to use MATLAB and LTspice
  - MOS design curves and trade-offs
  - Common source amplifier with resistive load
  - Common source amplifier with PMOS load

STRICTLY CONFIDENTIAL

THE TA TEAM



Arno Hemelhof
Master's degree from VUB/BRUFACE (Belgium, 2022)
Works on designing robust and efficient mmWave power amplifiers for 6G applications
Speaks Dutch and English



Chang-Shao Shen
Master's degree from NTU (Taiwan, 2021)
Works on electronic and photonic circuit design of distributed radar transceivers
Speaks Chinese and English



Ezgi Saç
Master's degree from KULeuven (Belgium, 2024)
Works on the design of digital transmit front-end in III-V technology for 5/6G applications
Speaks Turkish and English

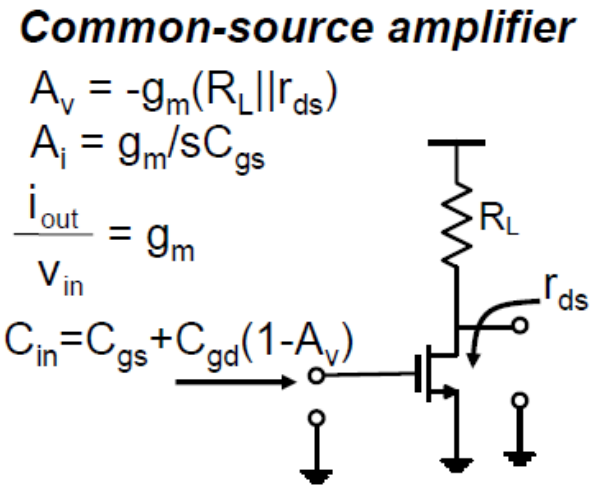


Xuepu Wu
Master's degree from ZJU (China, 2023)
Works on complex-sensing integrated circuits for built-in self-test of millimeter-wave transceivers
Speaks Chinese and English

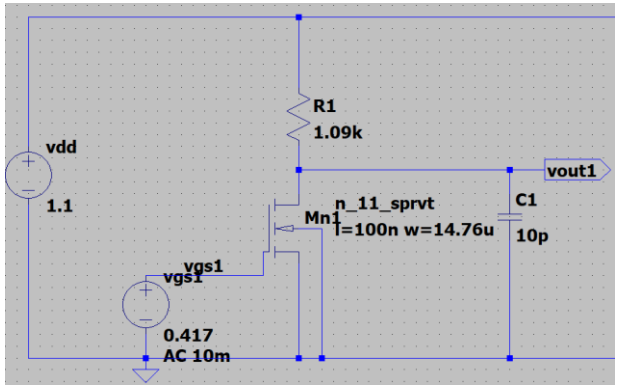
STRICTLY CONFIDENTIAL

CORE DIFFERENCES BETWEEN LECTURES AND LAB CLASSES

- Lectures:
  - Analysis of given architectures
  - Modeling of transistors and complex circuits
  - Finding equations for performance parameters (gain, bandwidth, power,...)



- Lab:
  - Quantitative design of the known architectures
  - How to size a transistor (width and length) to achieve a certain performance using Matlab
  - How to simulate a circuit using SPICE to validate the correctness of design



STRICTLY CONFIDENTIAL

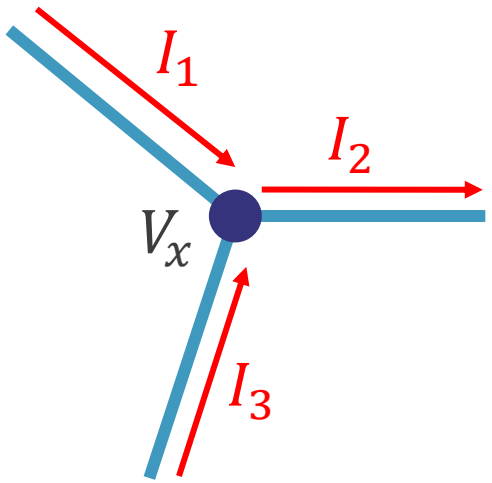
CONVENTION

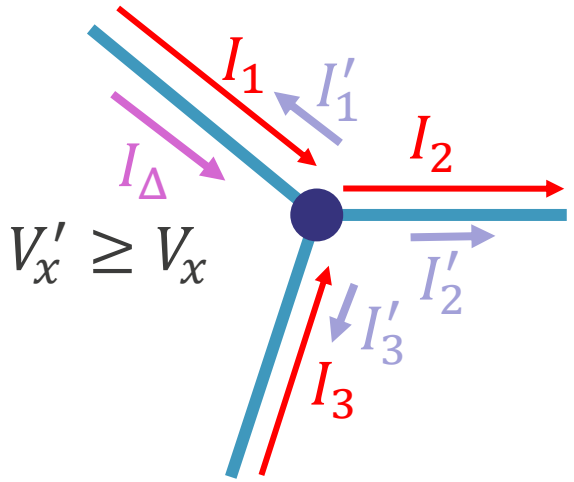
- DC: all uppercase (I_{DS}, V_{GS}, ...)
- Small-signal: all lowercase (i_{ds}, v_{gs}, ...)
- Large-signal: mixed case (I_{ds}, V_{gs}, ...)

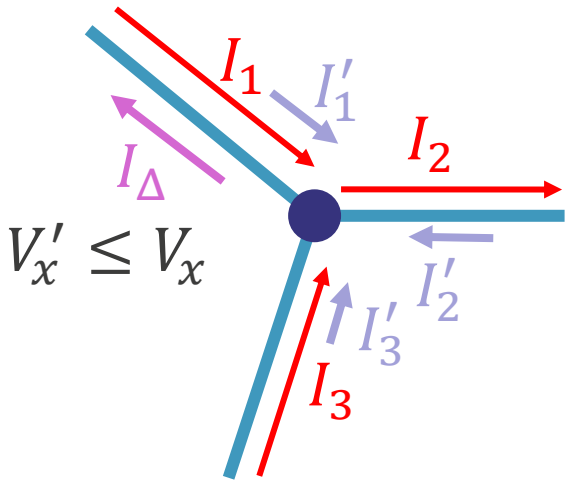
STRICTLY CONFIDENTIAL

PRIMER
REASONING ABOUT CIRCUITS

Kirchhoff's Current Law (KCL): $I_1 - I_2 + I_3 = 0$







Injecting current into a node increases the voltage

Extracting current from a node decreases the voltage

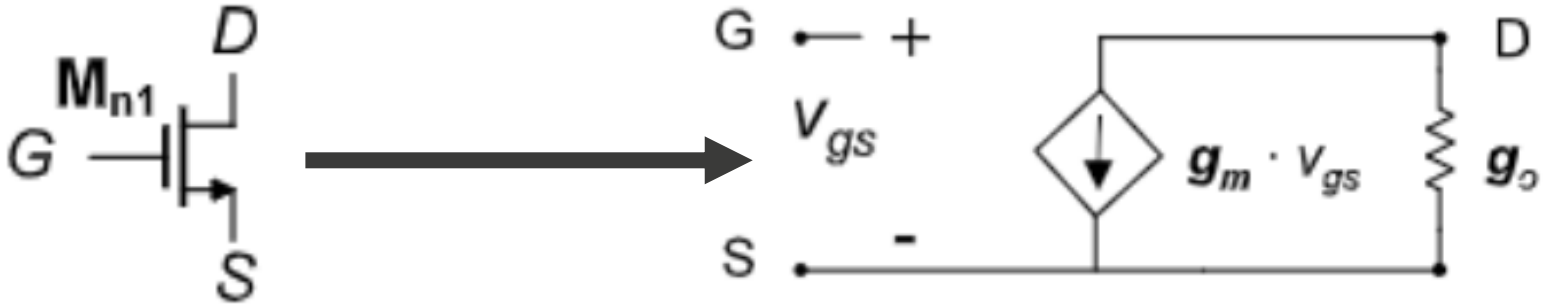
$$|\Delta V_x| = I_\Delta \cdot R_x$$

R_x is the equivalent resistance seen at node x (including the branch the branch containing I_Δ)

STRICTLY CONFIDENTIAL

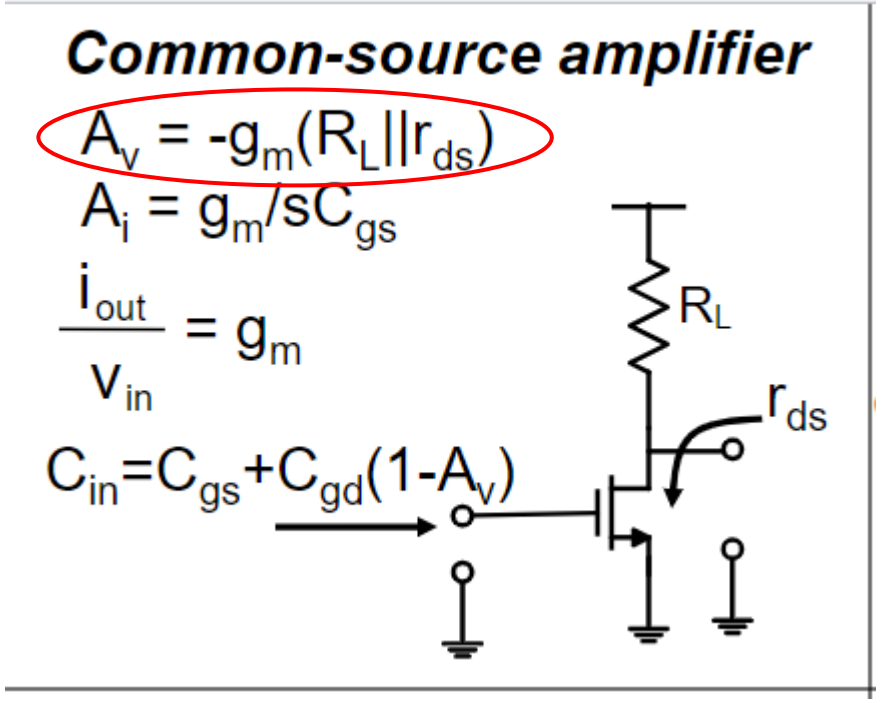
PRIMER

REASONING ABOUT CIRCUITS



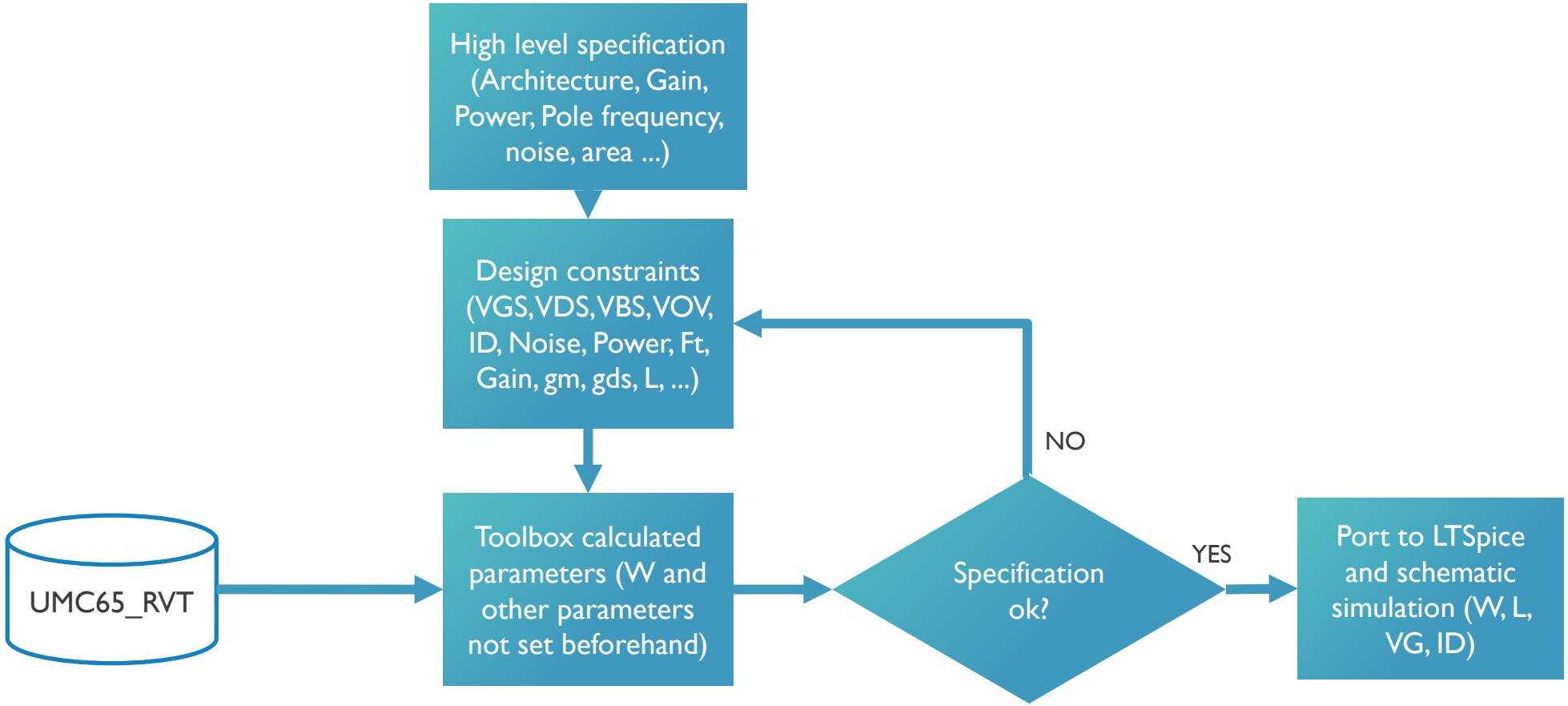
Also knows as g_{ds}

Increasing v_{gs} will increase i_{ds}, injecting current into **S** and extracting current from **D**



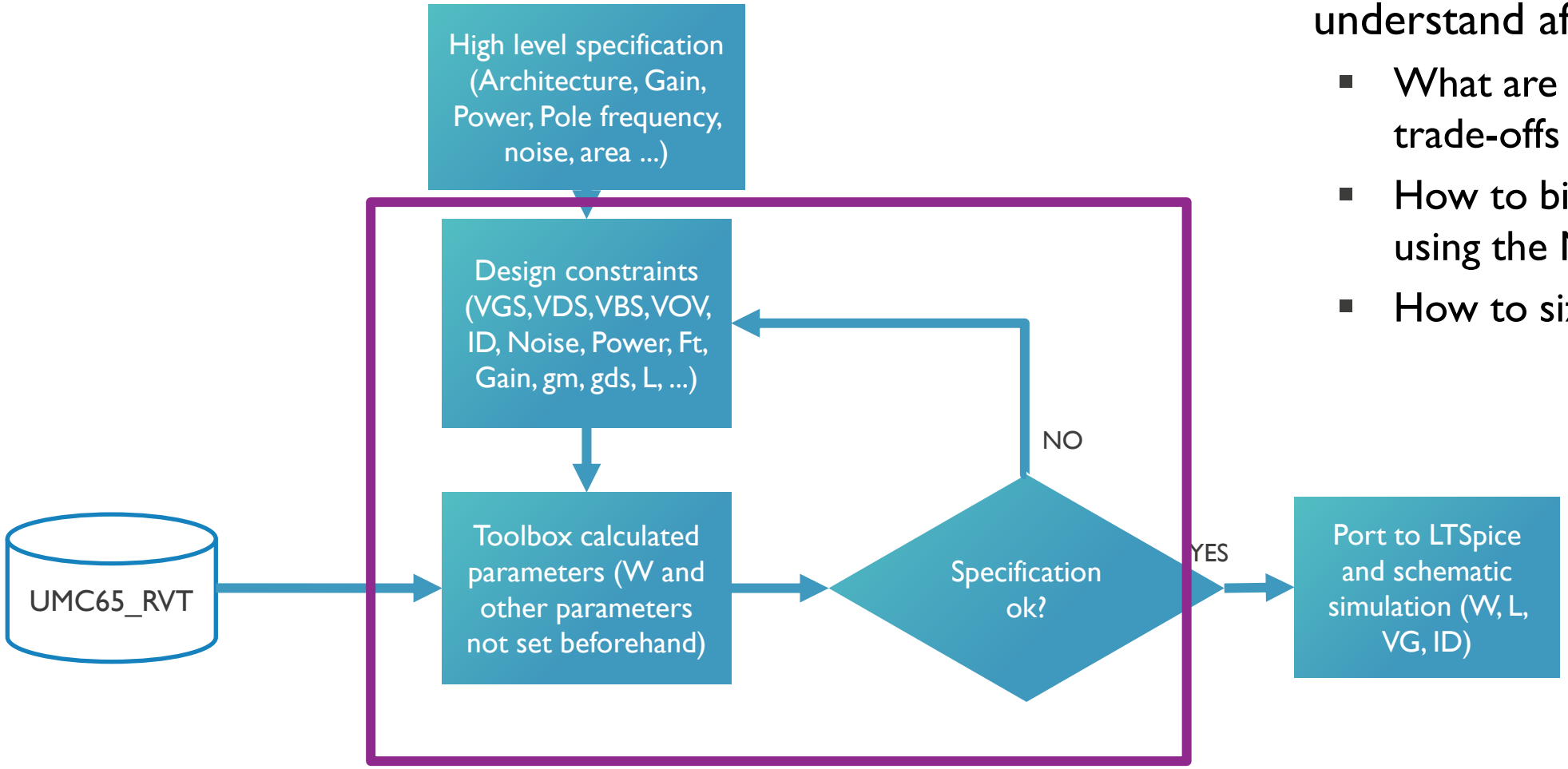
STRICTLY CONFIDENTIAL

HOW DOES THE DESIGN PROCESS WORK



STRICTLY CONFIDENTIAL

HOW DOES THE DESIGN PROCESS WORK



- What do you need to understand after this class?
  - What are the basic MOS trade-offs for design
  - How to bias a transistor using the MATLAB toolbox
  - How to size a transistor

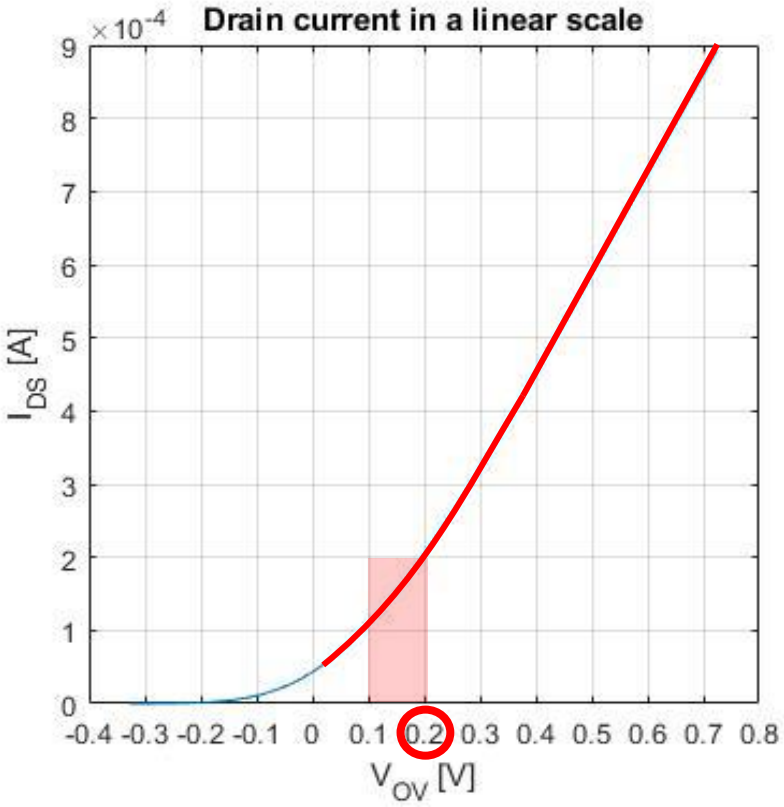
STRICTLY CONFIDENTIAL

BUILDING SOME INTUITON

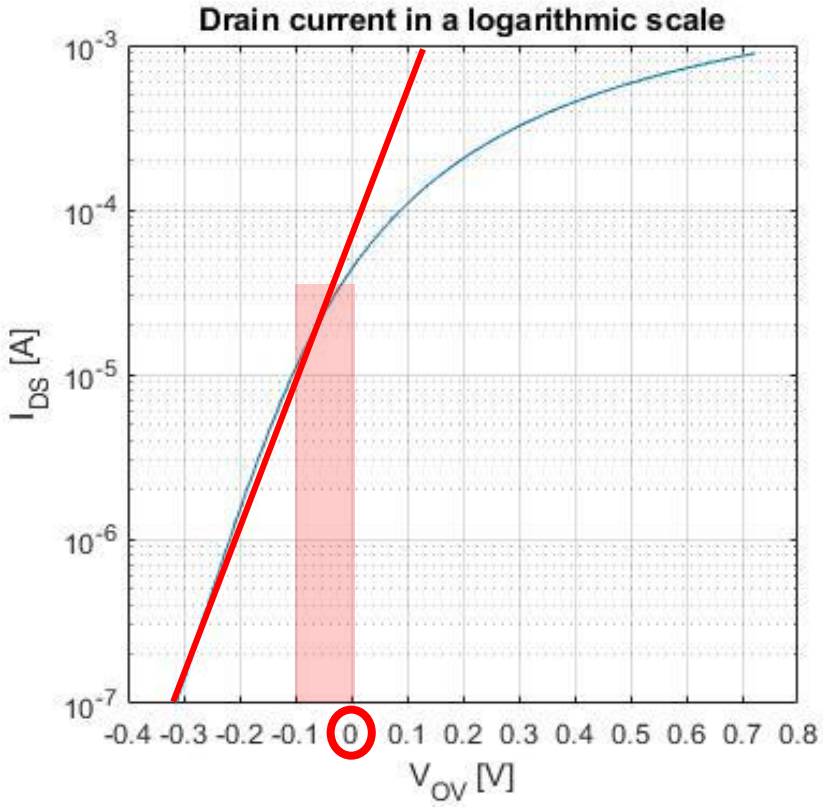
- Do exercises 1 and 2
  - ETA: 10 minutes
- Hints
  - Strong inversion: $I_{ds} \approx C_1 \cdot V_{ov}^2$ (or with velocity saturation $I_{ds} \approx C_2 \cdot V_{ov}$)
  - Weak inversion: $I_{ds} \approx C_3 \cdot \mathrm{e}^{\mathrm{Vov}} \rightarrow \ln I_{ds} \approx C_4 \cdot V_{ov} + C_5$
  - For exercise 2: there are many correct ways to obtain $1\ \mathrm{mS}$

STRICTLY CONFIDENTIAL

EXERCISE 1



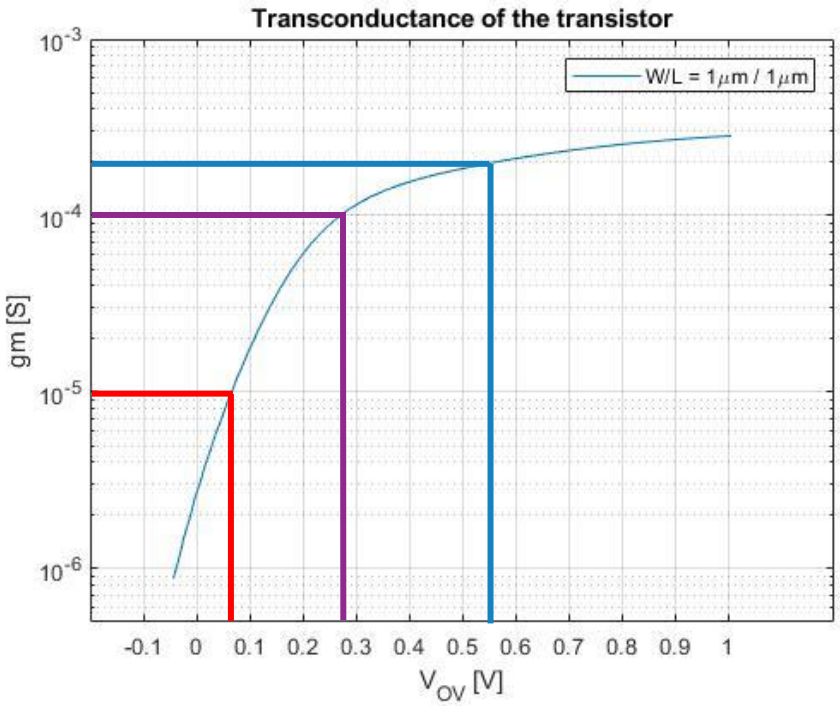
Quadratic in linear plot → Strong inversion
(linear due to velocity saturation)



Linear in log plot → Weak inversion

STRICTLY CONFIDENTIAL

EXERCISE 2



- 0.1mS
  - From graph: $V_{OV} = 0.3\text{V} \rightarrow V_{GS} = 0.6\text{V}$
- 1mS (multiple choices)
  - Take 10 times 0.1mS in parallel with $V_{GS} = 0.6\text{V}$
  - Take 5 times 0.2mS in parallel with $V_{GS} = 0.9V$
  - Take 100 times 0.01mS in parallel with $V_{GS} = 0.35V$

STRICTLY CONFIDENTIAL

USING THE MATLAB TOOLBOX

BOILERPLATE **Provided to you**

```
%% Adding paths + Loading MOS tables
addpath(genpath('circuitDesign'));
addpath(genpath('functions'));
addpath(genpath('models'));

w = warning ('off','all'); % Turn off warnings

clear;
close all;
clc;

load ('UMC65_RVT.mat');

%% Initialize everything
designkitName    = 'umc65';
circuitTitle     = 'Analog Design - Session 1';

%Declaration of the circuit components
elementList.nmos = {'Mn1','Mn2','Mn3','Mn4'};
elementList.pmos = {};

spec.VDD         = 1.1;
choice.maxFingerWidth = 10e-6;
choice.minFingerWidth = 200e-9;
simulator        ='spectre';
simulFile        = 0;
simulSkelFile    = 0;
analog = cirInit('analog', circuitTitle, 'top', elementList, spec , choice,...
    designkitName, NRVT, PRVT, simulator, simulFile, simulSkelFile);

analog           = cirCheckInChoice(analog, choice);

```

Loading functions and .mat file

Defining the transistors used in the circuit (Mn***X*** is a struct with the transistor parameters as fields)

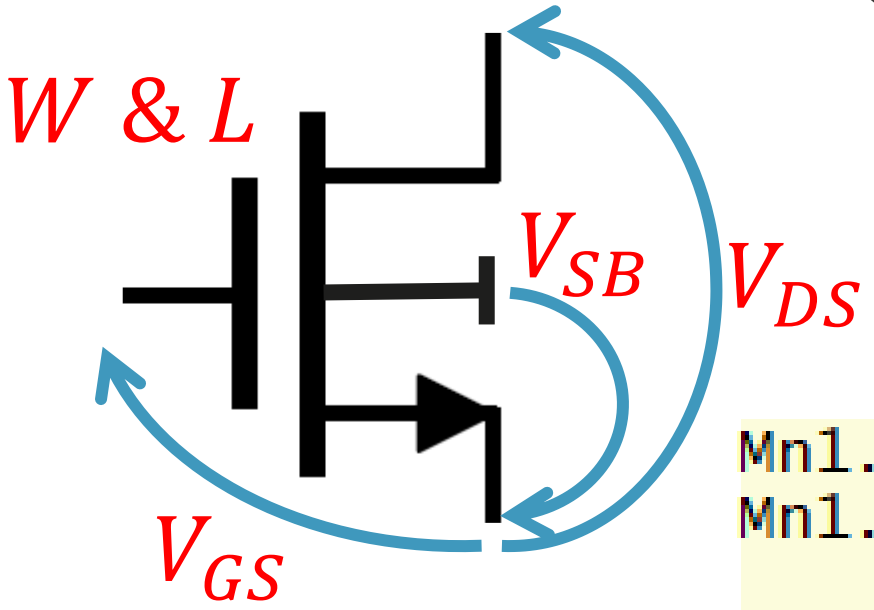
Defining technology constraints

Only change transistor list

STRICTLY CONFIDENTIAL

USING THE MATLAB TOOLBOX
SETTING THE DEVICE OPERATING POINT

1. Set values



All Required!
(S.I. Units)

```
Mn1.w = 10e-6;
Mn1.lg = 500e-9;

Mn1.vsb = 0;
Mn1.vgs = 0.4;
Mn1.vds = 0.55;
```

2. Cut fingers

```
Mn1 = mosNfingers(Mn1);
```

3. Calculate operating point

```
Mn1 = mosOpValues(Mn1);
```


STRICTLY CONFIDENTIAL

USING THE MATLAB TOOLBOX

GETTING V_{GS} FROM V_{OV}

$$V_{GS} = V_{OV} + V_{th}$$

NRVT: NMOS
PRVT: PMOS

V_{GS}
Doesn't Influence V_{th}

V_{SB}

```
Mn1.vth = tableValueWref('vth', NRVT, Mn1.lg, 0, Mn1.vds, Mn1.vsb);
```

Length

V_{DS}

STRICTLY CONFIDENTIAL

USING THE MATLAB TOOLBOX

GETTING W

If you know I_{DS}

```
Mn1.w    = mosWidth('ids', Mn1.ids, Mn1);
```

If you know g_m

```
Mn1.w    = mosWidth('gm', Mn1.gm, Mn1);
```


STRICTLY CONFIDENTIAL

PLOTTING THE MOS IDS CURVES
AGAINST VGS AND VDS

```
%% Circuit
disp('        Vd            ');
disp('        |             ');
disp('  Vg---Mnx            ');
disp('        |             ');
disp('        Vs            ');

fprintf('\n--- First Exercise: Designing transistor from scratch ---\n');
```

- **session1_part1.m**
- Fixed parameters: $W = 10\mu m, L = 65nm$
- Sweep parameters: VGS, VDS
- Calculated parameters: IDS (to be plot), VTH (byproduct of bias point) and other Operating point parameters (gm, gds, ...)

STRICTLY CONFIDENTIAL

WRITING THE CODE TOGETHER

USEFUL FUNCTIONS

tableValueWref

```
tableValueWref retrieving the value of an intrinsic MOS parameter for
the reference width, directly from a given table.

   RESULT = mosIntValueWref(PARAM, TABLE, LENGTH, VGS, VDS, VSB) returns
   the value of the MOS operating point parameter PARAM
   (specified as a string) for a given TABLE.
   PARAM must be exist as a field of the given TABLE.
   Possible names for PARAM can be found by running tableDisplay(TABLE).

   EXAMPLE :

      id = tableValueWref('ids', N, 0.18e-6, 0.7, vdd/2, 0)
```

- To extract the VTH value (weak dependency with MOS W)

mosNfingers

```
mosNfingers computation of the number of fingers for a MOS transistor.
```

- To slice the Width according to the maximum desired finger width

mosOpValues

```
mosOpValues computation of all operating parameters of a MOS transistor

   MOS = mosOpValues(MOS) computes the value of intrinsic and extrinsic
   operating point parameters (in S.I. units) of a given transistor MOS. The
   datastructure of the transistor which is specified as an argument is also
   returned. However, after this function has returned, all fields of the
   transistor related to intrinsic and extrinsic operating point parameters
   have been added to the datastructure of the transistor. Existing extrinsic
   geometry parameters are NOT recomputed. Also, the
   values of vgb, vgd and vdb are computed (or updated) from vgs, vds and vsb.

   EXAMPLE :

       Mn1 = mosOpValues(Mn1)
```

- To calculate all the other MOS parameters, in this case it is mainly Mn.ids

STRICTLY CONFIDENTIAL

ANALYSIS

- IDS graph
  - Reset to X-Z view: which regions can you see?
  - Reset to Y-Z view: which regions can you see?
- MOS saturation graph
  - Reset to Y-X view: can you discuss about the trends in Vdsat (Vds after which the transistor is in saturation)?

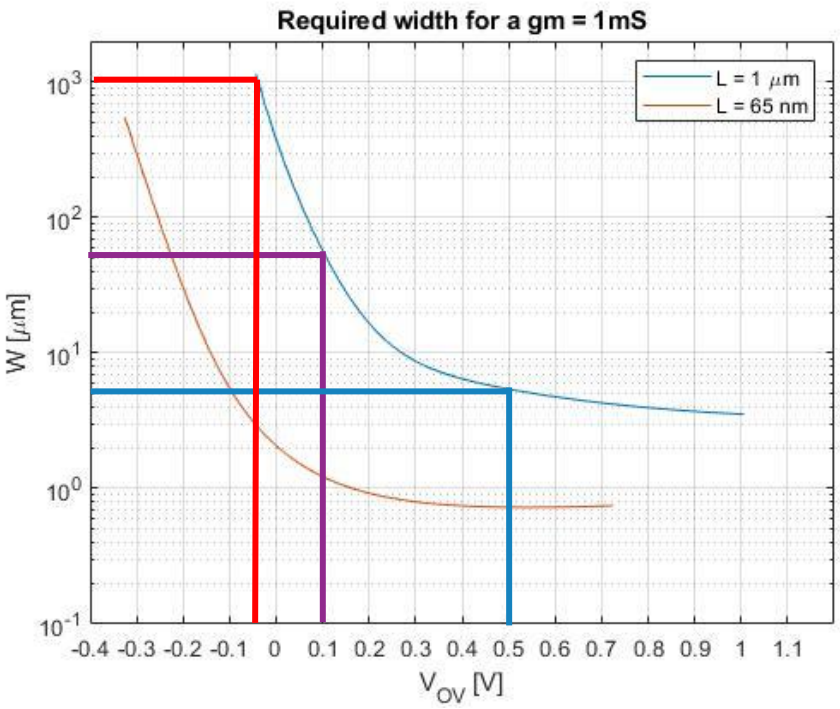
STRICTLY CONFIDENTIAL

BUILDING SOME INTUITION

- Exercise 3 and 4
  - ETA: 15 minutes
- In order to match results with the solution, use:
  - Weak inversion: $V_{OV} = -0.05\text{V}$
  - Moderate inversion: $V_{OV} = 0.1\text{V}$
  - Strong inversion: $V_{OV} = 0.5\text{V}$

STRICTLY CONFIDENTIAL

EXERCISE 3



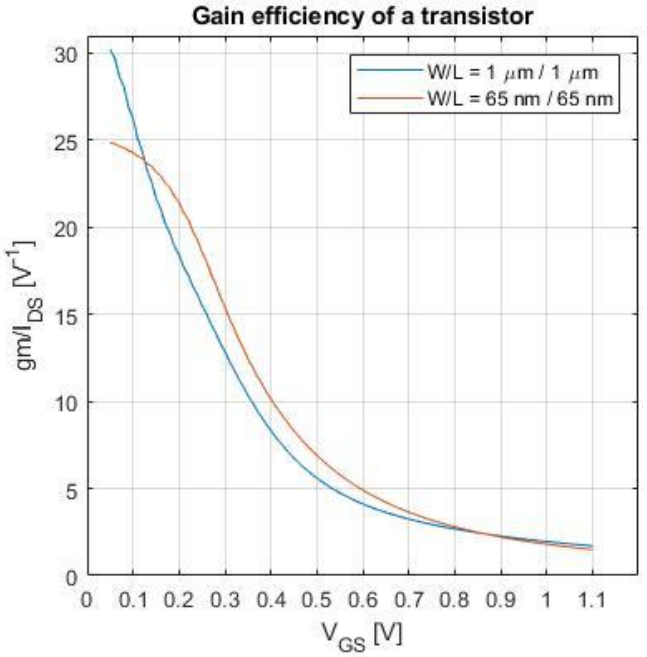
$$g_m = \mathrm{W} \cdot f(\text{other parameters})$$

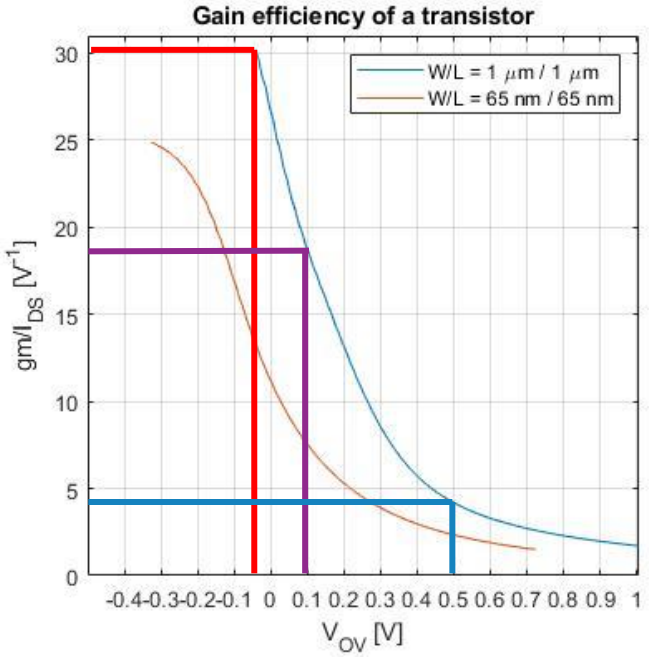
$$\Rightarrow \frac{g_{m1}}{g_{m2}} = \frac{W_1}{W_2} \text{ (if only } W \text{ changes)}$$

- Weak Inversion: $V_{OV} = -0.05V$
  - $\mathrm{W} = 1000\mu\mathrm{m}$ @ $1\mathrm{mS} \rightarrow \mathrm{W} = 10'000\mu\mathrm{m}$ @ $10\mathrm{mS}$
- Moderate Inversion: $V_{OV} = 0.1V$
  - $\mathrm{W} = 50\mu\mathrm{m}$ @ $1\mathrm{mS} \rightarrow \mathrm{W} = 500\mu\mathrm{m}$ @ $10\mathrm{mS}$
- Strong Inversion: $V_{OV} = 0.5V$
  - $\mathrm{W} = 5\mu\mathrm{m}$ @ $1\mathrm{mS} \rightarrow \mathrm{W} = 50\mu\mathrm{m}$ @ $10\mathrm{mS}$

STRICTLY CONFIDENTIAL

EXERCISE 4A





- Weak Inversion: $V_{OV} = -0.05V$
  - $\frac{g_m}{I_{DS}} = 31 \rightarrow I_{DS} = 322\mu A$
- Moderate Inversion: $V_{OV} = 0.1V$
  - $\frac{g_m}{I_{DS}} = 18 \rightarrow I_{DS} = 555\mu A$
- Strong Inversion: $V_{OV} = 0.5V$
  - $\frac{g_m}{I_{DS}} = 4 \rightarrow I_{DS} = 2500\mu A$

STRICTLY CONFIDENTIAL

EXERCISE 4B

- Gain $A_V = g_m \cdot R_L$
- $V_{DS} = V_{DD} - R_L \cdot I_{DS}$
- Examples (assuming an overdrive voltage of 0.1V)
  - $V_{DS} = 1.0\text{V} \rightarrow R_L = \frac{1.1\text{V} - 1.0\text{V}}{555\mu\text{A}} = 180\Omega \rightarrow A_V = 1.8\frac{\text{V}}{\text{V}}$
  - $V_{DS} = 0.55\text{V} \rightarrow R_L = \frac{1.1\text{V} - 0.55\text{V}}{555\mu\text{A}} = 990\Omega \rightarrow A_V = 9.9\frac{\text{V}}{\text{V}}$
  - $V_{DS} = 0.2\text{V} \rightarrow R_L = \frac{1.1\text{V} - 0.2\text{V}}{555\mu\text{A}} = 1.62\text{k}\Omega \rightarrow A_V = 16.2\frac{\text{V}}{\text{V}}$
- General: $A_V = g_m \cdot R_L = g_m \cdot \frac{V_{DD} - V_{DS}}{I_{DS}} = \frac{g_m}{I_{DS}} \cdot (V_{DD} - V_{DS})$
  - Smaller drain voltage leads to higher gain, but at what cost?

STRICTLY CONFIDENTIAL

COMPARISON TABLE

Implementing $g_m = 10\text{mS}$	W ($\propto$ Area)	I_{DS} ($\propto$ Power)	$A_V = g_m \cdot R_L$ @ $V_{DS} = 0.55\text{V}$
Weak Inversion ($V_{OV} = -0.05\text{V}$)	$10'000\mu\text{m}$	$370\mu\text{A}$	17.0 V/V (24.6dB)
Moderate Inversion ($V_{OV} = 0.1\text{V}$)	$500\mu\text{m}$	$714\mu\text{A}$	9.9 V/V (19.9dB)
Strong Inversion ($V_{OV} = 0.5\text{V}$)	$50\mu\text{m}$	$1100\mu\text{A}$	2.2 V/V (6.8dB)

STRICTLY CONFIDENTIAL

PLOTTING DEPENDENCY TO CHANNEL LENGTH

EXTRACTING SMALL SIGNAL PARAMETERS

```
%% Circuit
disp('     spec.VDD/2        ');
disp('        |              ');
disp('        |              ');
disp('  Vg---Mnx             ');
disp('        |              ');
disp('        Vs             ');

fprintf('\n--- First Exercise: Designing transistor from scratch ---\n');
```

- **session1_part2.m**
- Fixed parameters: VDS, W/L
- Sweep parameters: L, VGS
- Calculated parameters: g_m, g_{ds}, g_m/I_{DS} (gain efficiency), g_m/g_{ds} (intrinsic gain)

STRICTLY CONFIDENTIAL

ANALYSIS

- From an intrinsic gain perspective, which one is better?
  - Short channel or long channel?
  - Weak inversion or strong inversion?
- From a (gain) efficiency perspective, which one is better?
  - Short channel or long channel?
  - Weak inversion or strong inversion?

STRICTLY CONFIDENTIAL

HOMEWORK

- Extract the cutoff frequency (Mn.ft) value and plot alongside the previous graphs
  - For higher ft which length and inversion region are preferable?
- Extract the width calculated within the previous code and plot alongside the previous graphs
  - Which length and inversion level are preferable given a minimized area specification?

STRICTLY CONFIDENTIAL

RESISTIVE LOAD COMMON SOURCE STAGE

```
%% EX1: Circuit
disp('       VDD           ');
disp('        |            ');
disp('        RL           ');
disp('        |----+-OUT   ');
disp('  IN---Mn1   |       ');
disp('        |    CL      ');
disp('        |    |       ');
disp('       GND  GND      ');
```

- **session1_part3.m**
- Specification: Gain (>18dB), GBW (>100MHz) and Load (1pF) (given by system level decisions)
- Design choices: L, inversion region (VOV = VGS-VTH)
- Useful equations: $gm = 2\pi \times GBW \times C_{load}$ (note that gm turns to be a specification), $A_v = \frac{g_m}{g_{ds}+1/R_L}$

STRICTLY CONFIDENTIAL

WRITING THE MATLAB CODE

1. GBW = 100MHz, CL = 1pF, Gain > 18dB, Width < 1mm
2. Calculate target gm from CL and GBW
3. Set your design choices (Mn.lg, Mn.vov)
4. Set the terminal voltages (Mn.vds, Mn.vsb, VD = VDD/2)
5. Extract VTH
6. Set gate voltage (Mn.vgs = VOV+VTH)
7. Calculate width to realize given gm (Mn.w = mosWidth('gm', spec.gm, Mn))
8. Extract IDS from mosOpValues
9. Calculate RL from IDS and VD
10. Calculate gain (equation in previous slide)
    - If the specification was met, the design is ready to go to LTspice, if not go back to your design choices (lg and VOV) and start again reasoning towards which direction you want to go (long x short channel; weak x strong inversion)

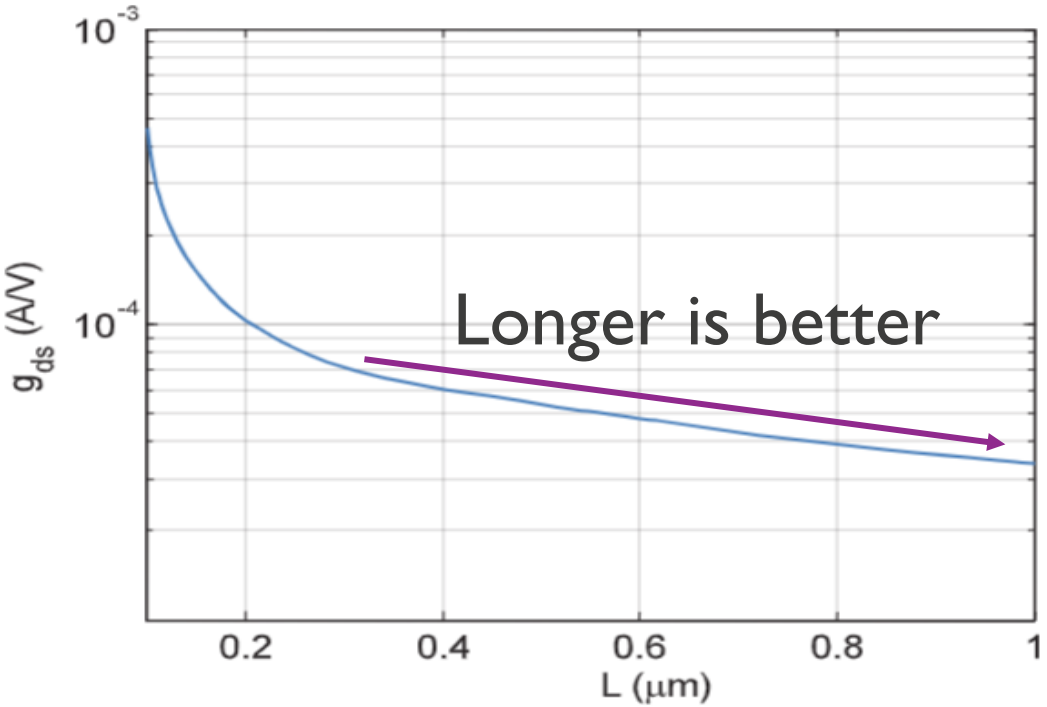
STRICTLY CONFIDENTIAL

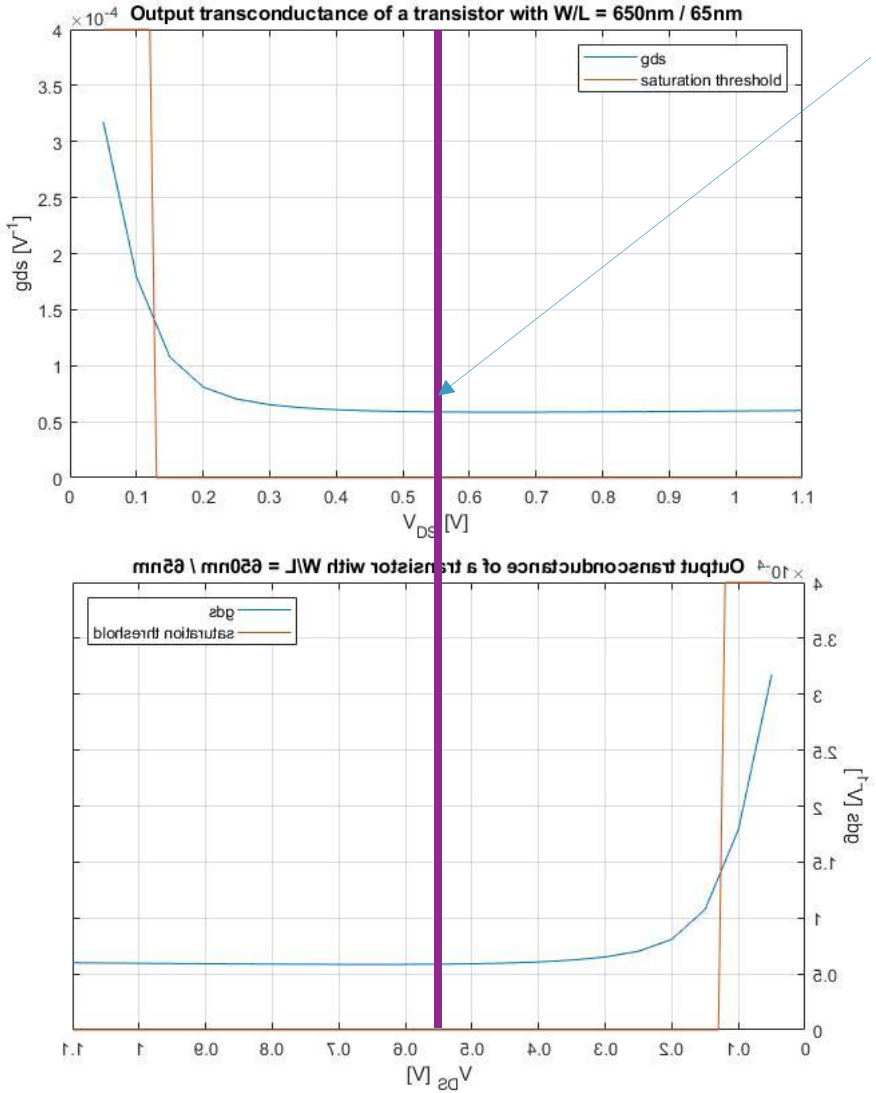
BUILDING SOME INTUITION

- Exercise 5
  - ETA: 15 minutes

STRICTLY CONFIDENTIAL

EXERCISE 5

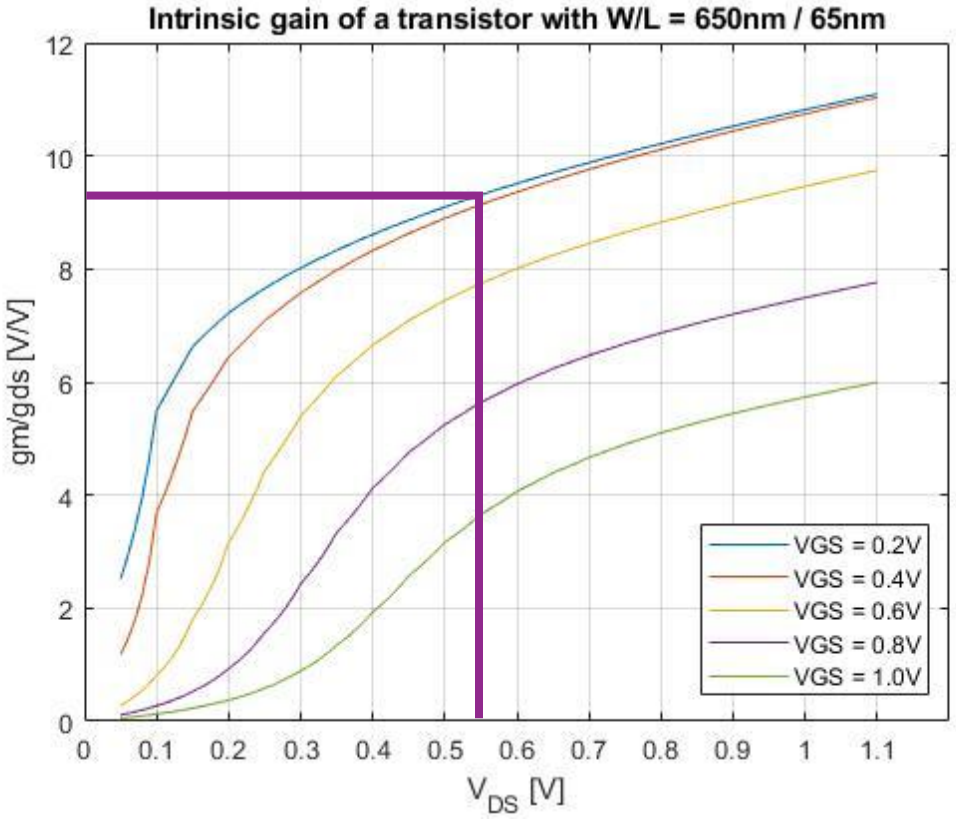




$V_{DS} = 0.55\text{V}$

EXERCISE 5

- $A_V = \frac{g_{m,\mathrm{NMOS}}}{g_{ds,\mathrm{NMOS}} + g_{ds,\mathrm{PMOS}}}$
- $g_{ds,\mathrm{PMOS}}$ is already minimized
  - Maximize $\frac{g_{m,\mathrm{NMOS}}}{g_{ds,\mathrm{NMOS}}}$



STRICTLY CONFIDENTIAL

PMOS LOAD COMMON SOURCE STAGE

```
%% EX2: Circuit
disp('       VDD           ');
disp('        |            ');
disp('       Mp2           ');
disp('        |----+-OUT    ');
disp('  IN---Mn2   |        ');
disp('        |    CL       ');
disp('        |    |        ');
disp('       GND  GND       ');
```

- Now there are two choices of L and VOV. What is the reasoning behind the first guess for length? And for the first guess of inversion level?
- Gain definition: $A_v = \frac{g_m^N}{g_{ds}^N + g_{ds}^P}$

STRICTLY CONFIDENTIAL

WRITING THE MATLAB CODE

1. GBW = 100MHz, CL = 1pF, **Gain > 26dB**, Width < 1mm
2. NMOS design has the same process as the case before
3. Sizing the PMOS load has some hard constraints
    1. Mp.vds + Mn.vds = spec.vdd (forced by supply)
    2. Mp.ids = Mn.ids (to provide a load functionality)
4. PMOS width is calculated based on its required IDS (Mp.ids = mosWidth('ids', Mp.ids, Mp))

STRICTLY CONFIDENTIAL